Active Directory Roaming Profiles

Setup and Configuration Guide

Secure, per-user roaming profile deployment for domain clients

Document ID	AD-SOP-004
Scope	Windows Server (AD DS) domain environments, Windows 10/11 clients
Owner	IT Specialist — Systems / Infrastructure
Prerequisites	Domain Admin or delegated OU rights; access to ADUC and the file server
Related SOPs	GPO Backup & Restore, Onboarding/Offboarding SOP, File Server ACL Standards

Overview

This guide documents the end-to-end configuration of secure, private roaming user profiles in an Active Directory domain. It covers network share provisioning, NTFS permission hardening for per-user isolation, linking the profile path in AD, and a verification procedure to confirm that profile folders are inaccessible to other standard users.

Roaming profiles allow a user's desktop settings, application data, and files (Desktop, Documents, etc.) to follow them to any domain-joined workstation. Because the profile store holds personal data for every user on the domain, correct permissioning is critical — a misconfigured share can expose one user's files to another.

Step 1 — Configure Network Share Permissions

Share-level permissions define the outer access boundary before NTFS permissions are applied. Use Advanced Sharing rather than the simple sharing wizard so that permissions can be set explicitly.

Principal	Share Permission
Authenticated Users	Change, Read
Administrators	Full Control

- Create the share (e.g. RoamingProfiles$) with the $ suffix to hide it from casual network browsing.
- Share permissions are intentionally permissive here (Change/Read) — the real access control is enforced at the NTFS layer in Step 2.

Step 2 — Configure Advanced NTFS Security Permissions

This is the step that actually enforces per-user privacy. Navigate to the Security tab of the share's root folder, open Advanced, and select Disable Inheritance → "Convert inherited permissions into explicit permissions." This isolates the folder's ACL from its parent so each principal below can be configured independently.

Principal	Access / Permissions	Applies To	Purpose
SYSTEM	Full Control	This folder, subfolders and files	Required by the OS to manage profile files.
Administrators	Full Control	This folder, subfolders and files	Allows IT to manage/restore profiles when needed.

Principal	Access / Permissions	Applies To	Purpose
CREATOR OWNER	Full Control	Subfolders and files only	Grants the profile owner full rights to their own subfolder only, once created.
Authenticated Users	List Folder / Read Data, Create Folders / Append Data (root share only)	This folder only	Lets any domain user create their own profile folder at first logon. Does NOT grant access into other users' folders — isolation comes from disabled inheritance + explicit ACEs above.

NOTE: Remove any general-purpose 'Users' or 'Everyone' style entries (e.g. SERVER\Users) from the ACL. Leaving a broad group with read access at this level is the most common cause of users being able to browse each other's profile folders.

Step 3 — Link the Profile Path in Active Directory

Open Active Directory Users and Computers (ADUC), right-click the target user, open Properties → Profile tab, and enter the universal profile path using the %username% variable:

```
\\serverlab\RoamingProfiles$\%username%
```

On Apply, AD automatically resolves %username% to the user's actual sAMAccountName (e.g. j.acebuche), so the same path string can be applied in bulk across many accounts via ADUC multi-select or PowerShell.

Step 4 — Test and Verify

1. Log in with the target user account on a domain-joined client PC.
2. Create a test file on the Desktop or in Documents of the client session.
3. Sign out / log off the client PC to force the profile to synchronize and upload to the server.
4. On the server, open the share root (e.g. C:\AD-shares\RoamingProfiles$). The user's profile folder should now exist, suffixed with .V6 (the profile version marker used by Windows 10/11).
5. Attempt to open that folder using an Administrator account that is not the profile owner. Receiving "You don't currently have permission to access this folder" confirms the isolation is working as intended — Administrators can still take ownership if recovery is ever required, but do not have standing read access.

Additional Considerations

- **Profile size quotas:** Set a maximum roaming profile size via Group Policy (User Configuration → Administrative Templates → System → User Profiles → "Limit the size of the entire roaming user profile cache") to prevent uncontrolled growth on the file server.
- **Mandatory vs. roaming profiles:** A roaming profile (NTUSER.DAT) allows user changes to persist across logons. A mandatory profile (NTUSER.MAN) resets on every logoff — useful for kiosk-style or shared-role machines, but not appropriate for standard users who need personalization retained.
- **Slow-link / VPN behavior:** By default, Windows may skip downloading a roaming profile over a detected slow link (including some VPN connections) and load the local cached copy instead. If remote or VPN users report profile changes not syncing, review the "Configure slow link speed for roaming user profiles" GPO setting.
- **Folder Redirection as an alternative:** For large data sets (Documents, Desktop), consider pairing or replacing roaming profiles with Folder Redirection, which avoids re-uploading large files at every logoff and reduces logon/logoff times.
- **Backup scope:** Confirm the roaming profile share is included in the file server's backup job — profile loss without a backup means full loss of user desktop state, not just files.

Revision History

Version	Date	Change
1.0	2026-09-02	Initial SOP created from lab-tested configuration (ACEBUCHE.LOCAL environment).